

Bharath Muppasani

 Columbia, South Carolina

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Education

University of South Carolina

Doctor of Philosophy, Computer Science

Advisors: Dr. Biplav Srivastava & Dr. Vignesh Narayanan

Highlights: Published at NeurIPS, AAAI, IROS, ICAPS; Awards: Best Demo Paper Award - AAAI 2024

International Institute of Information Technology

Bachelor in Technology, Electronics and Communication

Columbia, SC, USA

Jan 2022–Present

Naya Raipur, Chhattisgarh, India

Aug 2016 - May 2020

Experience

Graduate Research Assistant

Artificial Intelligence Institute, University of South Carolina

Columbia, SC

Jan 2022–Present

- Conducting research on multi-agent planning systems in dynamic environments, integrating model based and learning based planning techniques for efficient coordination to enable resource-constrained robotic deployments.

Research Intern

Adobe Research, Bengaluru, India

May 2026–Aug 2026

Research: Building Persuasive Conversational Systems

- Conducting research on persuasive conversational systems that generate context-aware and adaptive responses for effective, goal-oriented interactions.

Indian Institute of Technology, Kharagpur, India

May 2018–Aug 2018

Research: Smart Home Automation

- Analyzed Samsung multi-resident smart-home data using time-, sensor-, and frequency-based segmentation and contributed to an unsupervised activity-labeling approach – recognition from Samsung for research excellence.

Research Interests and Skills

- **Research Interests:** Learning (Deep Reinforcement Learning, Large Language Models), Reasoning (Automated Planning, Multi-Agent Path Finding), Representation (Ontology-based Knowledge Modeling).
- **Proficient:** Python, PDDL, C, SQL, SPARQL, Machine Learning Frameworks

Selected Projects

Hybrid Information-Centric Framework for Multi-Agent Path Finding

Jan 2025–Present

Related Publication: IROS-26

- Recent work introduces HI-MAPF, a hybrid information-centric multi-agent planning framework that reduces inter-agent communication by up to $510\times$ while maintaining coordination performance in real-world robot deployments.

Knowledge Engineering for Planning and Multi-Agent Systems

May 2022–Present

Related Publications: ICAPS-23 PLATO; CODS-COMAD-24; AAAI-26; AAAI-MAKE-26

- Built a planning ontology to capture domains, problems, plans, planners, and action-ordering constraints; leveraged this knowledge to support planner selection and performance analysis; released open resources and documentation.
- Extended to multi-agent settings with *maPO*: a planner-agnostic schema modeling agents, conflicts, subplans/joint plans, and provenance; developed a log-to-graph pipeline and SPARQL queries to generate conflict, causal, and global-plan explanations in natural language.

Planning Strategies for Dynamic Opinion Networks

May 2023–May 2024

Related Publications: NeurIPS-24, AAAI-24

- Investigated intervention planning in dynamic opinion networks to identify influential nodes and control misinformation spread by developing a ranking-based labeling method and a reinforcement learning planner, identifying effective strategies for misinformation control.
- Conceptualized information spread as a numerical planning problem and built a network visualization tool for simulating information propagation and visualizing plans for targeted information spread.

Related Publications: ICAPS-24; IJCAI-23; NeurIPS-23 GenPlan

- Developed a comprehensive dataset for fine-tuning and evaluating the performance of various Large Language Model (LLM) architectures, including encoder-only, decoder-only, and encoder-decoder configurations in automated planning scenarios.

Selected Publications

C = Conference W = Workshop J = Journal

- [C-1] Bharath Muppasani, Risha Patel, Biplav Srivastava, Vignesh Narayanan. *HI-MAPF: Towards Resource-Efficient Multi-Agent Deployment*. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2026).
- [C-2] Bharath Muppasani, Ritirupa Dey, Biplav Srivastava, Vignesh Narayanan. *OMEGA: An Ontology-Driven Tool for Explaining Multi-Agent Path Finding*. AAAI Conference on Artificial Intelligence (AAAI) – Demo Track (2026).
- [C-3] Bharath Muppasani, Vignesh Narayanan, Biplav Srivastava, Michael N. Huhns. *Towards Effective Planning Strategies for Dynamic Opinion Networks*. Neural Information Processing Systems (NeurIPS) Main Track (2024).
- [C-4] Bharath Muppasani, Vignesh Narayanan, Biplav Srivastava, Michael N. Huhns. *Expressive and Flexible Simulation of Information Spread Strategies in Social Networks Using Planning*. AAAI Conference – Demo Track (2024),
 **Best Demo Award**.
- [C-5] Vishal Pallagani, Bharath Muppasani, Kaushik Roy, Francesco Fabiano, Andrea Loreggia, Keerthiram Murugesan, Biplav Srivastava, Francesca Rossi, Lior Horesh, Amit Sheth. *On the Prospects of Incorporating Large Language Models (LLMs) in Automated Planning and Scheduling (APS)*. International Conference on Automated Planning and Scheduling (ICAPS) (2024).
- [C-6] Bharath Muppasani, Vishal Pallagani, Biplav Srivastava. *Solving the Rubik's Cube with a PDDL Planner*. ICAPS – Demo Track (2024).
- [C-7] Bharath Muppasani, Vishal Pallagani, Kausik Lakkaraju, Shuge Lei, Biplav Srivastava, Brett Robertson, Andrea Hickerson, Vignesh Narayanan. *On Safe and Usable Chatbots for Promoting Voter Participation*. *AI Magazine*, 44(3), AAAI (2023).
- [C-8] Bharath Muppasani, Cheyyur Jaya Anand, Chinmayi Appajigowda, Biplav Srivastava, Lokesh Johri. *A Dataset and Baseline Approach for Identifying Usage States from Non-intrusive Power Sensing with MiDAS IoT-Based Sensors*. Innovative Applications of Artificial Intelligence (IAAI) Conference, AAAI (2023).
- [J-1] Bharath Muppasani, Nitin Gupta, Vishal Pallagani, Biplav Srivastava, Raghava Mutharaju, Michael N. Huhns, Vignesh Narayanan. *Building a Plan Ontology to Represent and Exploit Planning Knowledge and Its Applications*. *Discover Data*, 3(1), Article 55 (2025).
- [W-1] Bharath Muppasani, Ritirupa Dey, Biplav Srivastava, Vignesh Narayanan. *maPO: An Ontology for Multi-Agent Path Finding and Its Usage for Explaining Planner Behaviour*. AAAI Workshop on Machine Learning and Knowledge Engineering for Knowledge-Grounded Semantic Agents (AAAI-MAKE), AAAI (2026).
- [W-2] Bharath Muppasani, Protik Nag, Biplav Srivastava, Vignesh Narayanan. *On Generalized Planning for Controlling Opinion Networks: Interpreting Human-AI Dialog States and Beliefs*. Generalization in Planning (GenPlan) Workshop, AAAI (2025).
- [W-3] Bharath Muppasani, Vishal Pallagani, Biplav Srivastava, Forest Agostinelli. *Comparing Rubik's Cube Solvability in Domain-Independent Planners Using Standard Planning Representations for Insights and Synergy with Upcoming Learning Methods*. Heuristics and Search for Domain-Independent Planning (HSDIP) Workshop, ICAPS (2024).
- [W-4] Kausik Lakkaraju, Sai Krishna Revanth Vuruma, Vishal Pallagani, Bharath Muppasani, Biplav Srivastava. *Can LLMs be Good Financial Advisors?: An Initial Study in Personal Decision Making for Optimized Outcomes*. FinPlan Workshop, ICAPS (2023).

Patents

- **Improving Planner Performance by Learning and Using Metadata of Experiences**

Accepted on March 24, 2023

Patent Office: United States (US)

Inventors: Vishal Pallagani, **Bharath Muppasani**, Biplav Srivastava